

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering
University Backlog Examination May 2011.
CE-204 Digital Electronics

Date: 19/05/11

Time: 10.00 a.m. To 1.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

Q-1 Attempt the following.

- | | | |
|---|---|----|
| 1 | What is AOI logic? | 01 |
| 2 | Add $(367)_8$ and $(715)_8$ in the given base without converting it to decimal. | 01 |
| 3 | How many cells are there on an n variable K-map? | 01 |
| 4 | Convert $(11010111.110)_2$ to decimal, octal and hexadecimal. | 01 |
| 5 | Using 10's complement perform $(72532)_{10} - (3250)_{10}$ | 01 |
| 6 | Represent $(250.5)_{10}$ in to base 4 number system. | 01 |
| 7 | Represent decimal number 8620 in Excess-3 code and 2 4 2 1 code. | 01 |

- Q-2(a) Find the complement of the function: $F1 = x(y'z' + yz)$ 04
 $F2 = x'yz' + x'y'z$
- (b) Differentiate between ripple counter and synchronous counter. 04
- (c) Determine the prime implicants of the function 06
 $F(w, x, y, z) = \sum m(1, 4, 6, 7, 8, 9, 10, 11, 15)$

OR

- Q-2 (a) Simplify the Boolean function using Boolean algebra. 06
(i) $A + B[AC + (B + C')D]$
(ii) Show that $AB + AB'C + BC' = AC + BC'$
- (b) What is race around condition in JK Flip flop? 03
- (c) Design a combinational circuit with four input lines that represent decimal digit in BCD and four output lines that generate the 9's complement of input Digit. 05
- Q-3(a) Implement full adder with a decoder and two OR gate. 03
- (b) Design 4 bit odd parity checker circuit. 05
- (c) Explain the operation of RS flipflop along with its symbol, characteristic table and characteristic equation. 06

OR

- Q-3 (a) Write a short note on Look ahead carry generator. 04
 (b) Explain the operation of edge trigger flipflop. 06
 (c) Built the following function in SOP and POS form using kmap. 04

$$F(A,B,C,D) = \sum(0,1,2,5,8,9,10)$$

SECTION II

- Q-4(a) Why a flipflop is called a single bit register? What does the 'state' of a flipflop indicate? 04
 (b) Convert the following function to other canonical form. 02
 (i) $F(A, B, C, D) = \sum(0, 2, 7, 12, 13, 14)$
 (ii) $F(x, y, z) = \prod(0, 2, 5, 6)$
 (c) What do you mean by K- map? 01
- Q-5 (a) Implement Full subtractor with two half subtractor and an OR gate 05
 (b) Explain operation of Binary down counter. 06
 (c) Describe the operation performed by the following logic circuits: 03
 (a)Decoder(b)Encoder(c)Comparator

OR

- Q-5(a) Implement the following function with NAND gate only. 04
 $F=AB+CD+E.$
 (b) What is shift register? What are the applications of shift register? 05
 (c) Write a short note on demultiplexer. 05
- Q-6(a) Write Short note on Johnson Counter. 06
 (b) Design a counter with the sequence 0,1,2,3,5,6 and repeat. Use JK Flip-flop. 06
 (c) Give the difference between combinational circuit and sequential circuit. 02

OR

- Q-6 (a) Implement the following function with multiplexer: 04

$$F(A,B,C,D)= \sum(0,1,3,4,8,9,15).$$

 (b) Explain serial transfer operation from register A to register B. 06
 (c) (a)Define the following: (1) State (2) State table. 04